# Compact Cutting-Plane for Congestion Games

Implementation Specification

## 1 Congestion Game Model

### 1.1 Network and players

Consider a directed graph $G = (V, E)$ with $|E| = K$ edges. There are $n$ agents. Each agent $i$ has a source $s_i \in V$ and a destination $t_i \in V$. Agent $i$'s action set $A_i$ is a set of $s_i$–$t_i$ paths. We write $m_i = |A_i|$.

Each action $a_i \in A_i$ is a path, represented as a subset of edges $a_i \subseteq E$.

### 1.2 Edge cost model

Each edge $e \in E$ has a latency function parameterized by $\theta_e \in \mathbb{R}^p$. The latency on edge $e$ depends on the congestion $\ell_e(a)$ (number of agents using edge $e$ under action profile $a$):

$$c_e\big(\ell_e(a)\big) \;=\; \sum_{j=1}^{p} \theta_{e,j} \cdot \ell_e(a)^{j-1} \;=\; \theta_e^\top \psi\big(\ell_e(a)\big)$$

where $\psi(\ell) = (1, \ell, \ell^2, \ldots, \ell^{p-1})^\top \in \mathbb{R}^p$ is the polynomial basis.

**Remark 1.** *For affine latencies ($p = 2$): $c_e(\ell) = \theta_{e,1} + \theta_{e,2}\,\ell$, so $\theta_e = (\theta_{e,1}, \theta_{e,2})$.*

### 1.3 Utility function

Agent $i$'s utility at action profile $a = (a_1, \ldots, a_n)$ is the negative total latency on their path:

$$u_i(a) \;=\; -\sum_{e \in a_i} c_e\big(\ell_e(a)\big) \;=\; -\sum_{e \in a_i} \theta_e^\top \psi\big(\ell_e(a)\big)$$

### 1.4 Parameter vector

Stack all edge parameters into a single vector:

$$\theta \;=\; (\theta_1^\top, \theta_2^\top, \ldots, \theta_K^\top)^\top \;\in\; \mathbb{R}^d, \qquad d = Kp$$

The utility is linear in $\theta$:

$$u_i(a) \;=\; -\phi_i(a)^\top \theta$$

where $\phi_i(a) \in \mathbb{R}^d$ is the **feature vector** for agent $i$ at profile $a$:

$$\phi_i(a) \;=\; \Big(\underbrace{\mathbf{1}[e \in a_i] \cdot \psi(\ell_e(a))}_{\text{block for edge } e}\Big)_{e \in E} \;\in\; \mathbb{R}^d$$

Blocks corresponding to edges *not* in $a_i$'s path are zero.

### 1.5 Utility differences

The utility difference (the object the cutting-plane learns) is:

$$w_i(a_i, a_i', a_{-i}) \;=\; u_i(a_i', a_{-i}) - u_i(a_i, a_{-i}) \;=\; -\big(\phi_i(a_i', a_{-i}) - \phi_i(a_i, a_{-i})\big)^\top \theta$$

Define the **difference feature**:

$$\Delta\phi_i(a_i, a_i', a_{-i}) \;:=\; \phi_i(a_i, a_{-i}) - \phi_i(a_i', a_{-i}) \;\in\; \mathbb{R}^d$$

so that $w_i(a_i, a_i', a_{-i}) = \Delta\phi_i(a_i, a_i', a_{-i})^\top \theta$.

In the full normal-form representation, the packed $w$-vector satisfies $w = \Phi\,\theta$ where $\Phi \in \mathbb{R}^{N \times d}$ stacks all $\Delta\phi_i$ vectors. But we **never form $\Phi$ explicitly**.

## 2 Compact Correlated Equilibrium LP

### 2.1 Why the standard CE LP is too large

The standard CE LP has $M = \prod_i m_i$ variables (one per action profile). For $n = 20$ agents with $m_i = 3$ routes each, $M = 3^{20} \approx 3.5 \times 10^9$. Intractable.

### 2.2 Route-flow formulation

Instead of variables $x(a)$ for each profile $a \in A$, we use a **route-flow** formulation. Define:

$$f_{i,r} \;=\; \Pr_{a \sim x}[a_i = r] \qquad \text{for each agent } i,\ \text{route } r \in A_i$$

This is the marginal probability that agent $i$ plays route $r$. The number of variables is $\sum_i m_i$ (polynomial in $n$).

**Remark 2.** *The marginals $f_{i,r}$ do not determine the joint distribution $x$ uniquely. However, for congestion games, the CE constraints depend on $x$ only through certain **congestion moments** (expected congestion on edges), which can be expressed in terms of the marginals under a product or low-correlation approximation. For **exact** CE computation in congestion games, one can use the approach of Papadimitriou and Roughgarden (2008), which exploits the structure of congestion games to solve the CE LP in polynomial time.*

### 2.3 CE constraints in the congestion game

For agent $i$, recommended route $r \in A_i$, deviation route $r' \in A_i$, the CE incentive constraint is:

$$\sum_{a_{-i}} x(r, a_{-i})\big[u_i(r', a_{-i}) - u_i(r, a_{-i})\big] \;\le\; 0$$

Substituting the linear utility model:

$$\sum_{a_{-i}} x(r, a_{-i})\ \Delta\phi_i(r, r', a_{-i})^\top \theta \;\le\; 0$$

This can be written as:

$$\Big(\sum_{a_{-i}} x(r, a_{-i})\ \Delta\phi_i(r, r', a_{-i})\Big)^\top \theta \;\le\; 0$$

The vector in parentheses is the **expected difference feature** under the conditional distribution $x(\cdot \mid a_i = r)$.

### 2.4 Product-distribution CE approximation

When the joint distribution $x$ is a product of marginals $x(a) = \prod_i f_{i,a_i}$, the expected congestion on edge $e$ is:

$$\bar{\ell}_e \;=\; \sum_{i=1}^{n} \sum_{r \in A_i} f_{i,r}\, \mathbf{1}[e \in r]$$

and the expected difference feature can be computed from $\bar{\ell}_e$ and the marginals without enumerating all profiles.

**Compact LP variables:** $f_{i,r}$ for $i \in [n]$, $r \in A_i$. Total: $\sum_i m_i$.

**Compact LP constraints:**

- Simplex per agent: $\sum_{r \in A_i} f_{i,r} = 1,\ f_{i,r} \geq 0$.
- CE incentive: for each $i$, $r$, $r'$: expected gain from deviating $\leq 0$. Under product distribution, these involve $\bar{\ell}_e$ and are polynomial-size.

**Remark 3.** *The product-distribution approximation yields a **coarse correlated equilibrium** (CCE), not an exact CE. For exact CE, one needs the Papadimitriou-Roughgarden approach. For the experiments, CCE is a reasonable starting point and is much simpler to implement.*

## 3 Compact Cutting-Plane Pipeline

The full pipeline operates in $\mathbb{R}^d$ (the parameter space) instead of $\mathbb{R}^N$ (the normal-form utility space).

### 3.1 State

- $\theta(t) \in \mathbb{R}^d$: current estimate of edge cost parameters.
- Knowledge set $C_t = \{\theta \in [-R, R]^d : \langle q_\theta^{(s)}, \theta \rangle \geq 0,\ s = 1, \ldots, t\}$.
- Accumulated cuts $q_\theta^{(1)}, \ldots, q_\theta^{(t)} \in \mathbb{R}^d$.

### 3.2 Per-round operations

**Step 1: Query point.** Compute $\theta(t) =$ analytic center of $C_{t-1}$ via ACCPM in $\mathbb{R}^d$.

Cost: $\mathcal{O}(d \cdot t^2)$ per Newton step via Woodbury.

**Step 2: Reconstruct utilities.** From $\theta(t)$, compute the utility differences needed for the CE LP. For each agent $i$, route pair $(r, r')$, and edge $e$:

$$w_i(r, r')_e \;=\; \Delta\phi_i(r, r', \cdot)^\top \theta(t)$$

This requires evaluating $\Delta\phi_i$ for each $(i, r, r', e)$ but **not** for each opponent profile $a_{-i}$ separately.

Under the product-distribution CE, the expected congestion depends on the marginals, which are LP variables. So we embed $\theta(t)$ into the CE LP constraints directly.

Cost: $\mathcal{O}(n \cdot m_{\max}^2 \cdot K)$ where $m_{\max} = \max_i m_i$.

**Step 3: Solve compact CE LP.** Find $f = (f_{i,r})$ that forms a (coarse) correlated equilibrium of the game with parameters $\theta(t)$.

Number of variables: $\sum_i m_i$.

Number of constraints: $\sum_i m_i(m_i - 1)$ incentive constraints + $n$ simplex constraints.

Cost: Gurobi LP in $\sum_i m_i$ variables. Very fast.

**Step 4: Sample and recommend.** Sample a route $r_i \sim f_{i,\cdot}$ independently for each agent $i$. Send $r_i$ privately to agent $i$.

Cost: $\mathcal{O}(n)$.

**Step 5: Observe feedback.** Each agent $i$, receiving recommendation $r_i$ and knowing $f$, best-responds (or quantal-responds). The moderator observes the realized routes $r_i^*$ for all agents.

The realized action profile $a^* = (r_1^*, \ldots, r_n^*)$ determines the congestion $\ell_e(a^*)$ on each edge.

**Step 6: Construct cut in $\mathbb{R}^d$.** For each deviating agent $i$ ($r_i^* \neq r_i$):

The separation oracle (Algorithm 4 of the paper) produces a cut $q \in \mathbb{R}^N$. But we compute the projected cut directly in $\mathbb{R}^d$ without ever forming $q$:

$$q_\theta \;=\; \sum_{i:\, r_i^* \neq r_i} x(r_i, \cdot)^\top \Delta\Phi_i(r_i, r_i^*)$$

where $x(r_i, \cdot)$ is the conditional distribution over opponents given $r_i$, and $\Delta\Phi_i(r_i, r_i^*)$ stacks the difference features.

Under the product distribution, $x(r_i, a_{-i}) = \prod_{j \neq i} f_{j,a_j}$, so:

$$q_\theta \;=\; \sum_{i:\, r_i^* \neq r_i} \sum_{a_{-i}} \Big(\prod_{j \neq i} f_{j,a_j}\Big) \Delta\phi_i(r_i, r_i^*, a_{-i})$$

**Key simplification for congestion games.** The feature $\Delta\phi_i(r_i, r_i^*, a_{-i})$ depends on $a_{-i}$ only through the congestion $\ell_e(a_{-i})$ on edges in $r_i \cup r_i^*$. The expected congestion under the product distribution is:

$$\mathbb{E}[\ell_e(a_{-i})] \;=\; \sum_{j \neq i} \sum_{r \in A_j} f_{j,r}\, \mathbf{1}[e \in r]$$

For the polynomial basis $\psi(\ell) = (1, \ell, \ldots, \ell^{p-1})$, we need the moments $\mathbb{E}[\ell_e^k]$ for $k = 0, \ldots, p-1$. Under a product distribution, these moments can be computed in $\mathcal{O}(n)$ per edge via the Poisson binomial distribution.

For **affine latencies** ($p = 2$), we only need $\mathbb{E}[\ell_e]$ and $\mathbb{E}[\ell_e^2]$:

$$\mathbb{E}[\ell_e] = \sum_{j \neq i} p_{j,e}, \qquad p_{j,e} := \sum_{r \in A_j} f_{j,r}\, \mathbf{1}[e \in r]$$

$$\mathbb{E}[\ell_e^2] = \mathbb{E}[\ell_e]^2 + \sum_{j \neq i} p_{j,e}(1 - p_{j,e})$$

This gives us $q_\theta \in \mathbb{R}^d$ directly, without enumerating opponent profiles.

Cost: $\mathcal{O}(n \cdot K \cdot p)$.

**Step 7: Update knowledge set.** Add $q_\theta$ to the cut list. Compute next analytic center.

### 3.3 Separation property

The projected cut preserves the separation property:

$$\langle \theta^*, q_\theta \rangle = \langle \Phi\, \theta^*, q \rangle_{\mathbb{R}^N} = \sum_i \phi_i(r_i, r_i^*, x) \;\geq\; 0$$

$$\langle \theta(t), q_\theta \rangle = \langle \Phi\, \theta(t), q \rangle_{\mathbb{R}^N} \;\leq\; 0$$

where the first inequality holds because agents deviate to profitable actions, and the second because $x(t) \in \mathrm{CE}(\theta(t))$.

# 4 Complexity Summary

| Operation | Normal-form | Congestion (compact) |
|---|---|---|
| ACCPM (per Newton) | $\mathcal{O}(N \cdot t^2)$ | $\mathcal{O}(d \cdot t^2)$ |
| CE LP variables | $M = \prod_i m_i$ | $\sum_i m_i$ |
| CE LP constraints | $\sum_i m_i(m_i - 1)$ | $\sum_i m_i(m_i - 1)$ |
| Cut construction | $\mathcal{O}(M)$ | $\mathcal{O}(n \cdot K \cdot p)$ |
| $N$ (parameter dim) | $\sum_i (m_i - 1)\lvert A_{-i}\rvert$ | $d = Kp$ |

Table 1: Per-round cost comparison. $K$ = edges, $p$ = polynomial degree, $d = Kp$, $t$ = accumulated cuts, $M = \prod_i m_i$, $N$ = normal-form utility dimension.

**Example:** $n = 100$ agents on a network with $K = 50$ edges, affine latencies ($p = 2$), $m_i = 5$ routes each.

- Normal-form: $M = 5^{100} \approx 10^{70}$. **Completely intractable.**
- Compact: $d = 100$, CE LP has 500 variables. **Trivial.**

# 5 Implementation Plan

## 5.1 Module: `congestion_game.py`

1. **Network definition:** Graph $G = (V, E)$, source-destination pairs $(s_i, t_i)$, route sets $A_i$ (enumerate $s_i$–$t_i$ paths up to a length bound).
2. **Feature computation:** Given an action profile $a$ and agent $i$, compute $\phi_i(a) \in \mathbb{R}^d$ and $\Delta\phi_i(r, r', a_{-i}) \in \mathbb{R}^d$.
3. **Utility evaluation:** $u_i(a) = -\phi_i(a)^\top \theta$.
4. **Congestion computation:** Given marginals $f$, compute expected congestion moments $\mathbb{E}[\ell_e^k]$ per edge.

## 5.2 Module: `compact_ce_solver.py`

1. **Build LP:** Variables $f_{i,r} \geq 0$, simplex constraints, CE incentive constraints using $\theta(t)$ and expected congestion moments.
2. **Solve via Gurobi:** LP with $\sum_i m_i$ variables.
3. **Return marginals:** $f_{i,r}$ for each agent $i$ and route $r$.

## 5.3 Module: `compact_oracle.py`

1. **Sample:** Route $r_i \sim f_{i,\cdot}$ independently per agent.
2. **Feedback:** Each agent best-responds (or quantal-responds) to the recommendation given the marginals. Observe realized routes $r_i^*$.
3. **Cut in $\mathbb{R}^d$:** For each deviating agent, compute $q_\theta$ using expected congestion moments. No enumeration of $a_{-i}$.

## 5.4 Module: `compact_cutting_plane.py`

1. **Reuse existing ACCPM** from `cutting_plane.py`, but in $\mathbb{R}^d$ instead of $\mathbb{R}^N$.

2. **Per-round loop:** $\theta(t)$ from ACCPM $\to$ compact CE LP $\to$ sample $\to$ feedback $\to$ cut in $\mathbb{R}^d$ $\to$ update.
3. **Track regret:** Both platform compliance regret and player swap regret, same as before.

### 5.5 Experiment

Run the compact pipeline on a Braess-like network:

- $n = 10, 50, 100, 500$ agents.
- $K = 10$–$50$ edges, affine latencies ($p = 2$, so $d = 2K$).
- $m_i = 3$–$5$ routes per agent.
- Compare with the normal-form pipeline (for small $n$ where both are feasible).
- Show that regret scaling matches theory and that the compact pipeline handles $n = 500$ in seconds.